is often based on introspection, political considerations, and aesthetics; and they tend to be much more skeptical of the validity of any scientific (or mathematical, or logical) results than their analytic colleagues.

A. J. Ayer summed up the continuing influence of positivism among analytic philosophers well. In 1982, long after logical positivism of the sort the Vienna Circle advocated had been discarded by the philosophical community, Ayer wrote, "Few of the principal theses of the Vienna Circle survive intact.… [But] I think it can be said that the spirit of Viennese positivism survives. In its re-accommodation of philosophy with science, its logical techniques, its insistence on clarity, its banishment of what I can best describe as a strain of wooly uplift from philosophy, it gave a new direction to the subject which is not now likely to be reversed" (1982, pp. 140–141).

As Continental philosophers are generally in the minority among philosophers of physics, and as they have not contributed nearly as much to scientific discussions about the interpretation of quantum physics, I will be almost entirely ignoring them in this book. Elsewhere, where I refer to "philosophers," you should mentally annotate that as "analytic philosophers."

scientific realism: Kuhn, Feyerabend, and Hanson were not realists, but Smart, Putnam, Popper, Maxwell, and much of the rest of the philosophical community—including some former members of the Vienna Circle, like Herbert Feigl—were convinced by the arguments in favor of realism. Today, philosophers of physics are, by a wide majority, realists of some stripe.

"what we see through an ordinary windowpane": Grover Maxwell 1962, "The Ontological Status of Theoretical Entities." *Minnesota Studies in the Philosophy of Science* 3:7.

"separating the observable from the unobservable": Ibid., p. 11.

"no longer seem surprising": J. J. C. Smart 1963, *Philosophy and Scientific Realism* (Routledge and Kegan Paul), p. 39.

"if there really were a criminal": Ibid., p. 47.

"doesn't make the success of science a miracle": Hilary Putnam 1979, *Mathematics, Matter, and Method,* 2nd ed. (Cambridge University Press), p. 73.

"alternatives to the prevailing Copenhagen interpretation": Smart 1963,